Project: Worked examples
Title: Column buckling (Perry–Robertson)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Compressive strength and resistance of a pin-ended strut of a given effective length, from Euler's critical stress reduced for initial imperfection by the Perry–Robertson formula, with the Robertson constant chosen by strut curve; utilisation against an applied axial load.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Radius of gyration and slenderness | $r = \sqrt{(I/A)}$; $\lambda = L_E/r$ |
| Euler buckling stress and load | $p_E = \pi^2 E/\lambda^2$; $P_E = p_E \cdot A$ |
| Limiting slenderness, below which the strut is not reduced for buckling | $\lambda_0 = 0.2 \cdot \sqrt{(\pi^2 E/p_y)}$ |
| Perry factor (Robertson constant a = 2.0, 3.5, 5.5, 8.0 for curves a–d) | $\eta = \max(0.001 \cdot a \cdot (\lambda - \lambda_0), 0)$ |
| Compressive strength by the Perry–Robertson formula | $\varphi = (p_y + (\eta + 1) \cdot p_E)/2$; $p_c = p_E \cdot p_y/(\varphi + \sqrt{(\varphi^2 - p_E \cdot p_y)})$ |
| Compression resistance and utilisation | $P_c = p_c \cdot A$; $u = P/P_c$ |

Basis: Compressive strength (Perry strut formula)[1](Annex C); Maximum slenderness[1](4.7.3.2); Allocation of strut curve[1](Table 23).

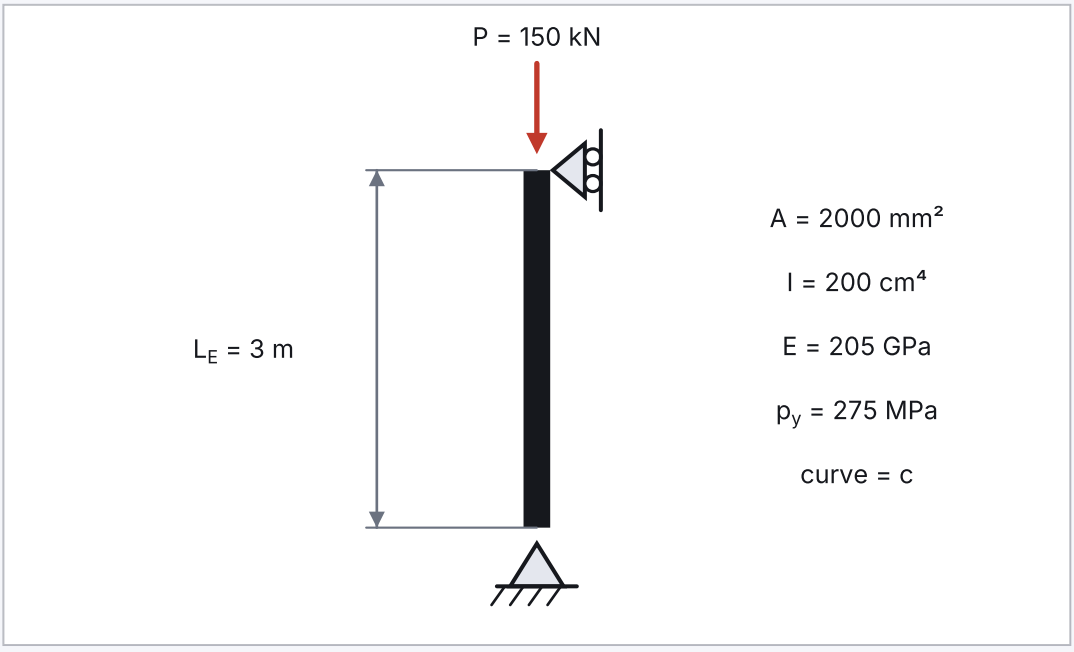



Figure 1 — A pin-ended strut under axial compression, effective length $L_E$. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $L_E$ | Effective length | 3 | m | drawing: DRG-ST-410 rev B | 0.1 … 50 m |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| A | Cross-section area | 2000 | $mm^2$ | spec: SPEC-ST-02 section schedule | 10 … 1000000 $mm^2$ |
| I | Second moment of area about the weaker axis | 200 | $cm^4$ | spec: SPEC-ST-02 section schedule | 0.01 … 10000000 $cm^4$ |
| E | Modulus of elasticity | 205 | GPa | spec: BS 5950-1 3.1.3: E = 205 GPa | 50 … 450 GPa |
| $p_y$ | Design strength (yield) | 275 | MPa | spec: SPEC-ST-02: S275, t ≤ 16 mm | 100 … 1000 MPa |
| curve | Strut curve (selects the Robertson constant) | c | | spec: BS 5950-1 Table 23: welded box, curve c | any |
| P | Applied axial load | 150 | kN | assumption: Factored axial load from the frame analysis | 0 … 100000 kN |

## 3. Assumptions

A1. Pin-ended, prismatic strut under axial load only; buckling about the axis of the given second moment. (Effective length carries the end restraint.)
A2. Initial imperfection represented by the Perry factor $\eta = 0.001 \cdot a \cdot (\lambda - \lambda_0)$, with a from the strut curve. (Robertson's fit to strut tests.)
A3. P (Applied axial load) is taken as 150 kN. (Factored axial load from the frame analysis)

## 4. Working

[8] $a_R = 5.5$ [1](Annex C)
= 5.5
Robertson constant for strut curve c
[9] r = sqrt(I / A)
= sqrt(200 $cm^4$ / 2000 $mm^2$)
= 31.623 mm
Radius of gyration
[10] $\lambda = L_E$ / r
= 3 m / 31.623 mm
= 94.868
Slenderness
[11] $p_E$ = pi^2 * E / λ^2 [1](Annex C)
= π^2 × 205 GPa / 94.868^2
= 224.81 MPa
Euler critical stress
[12] $P_E = p_E$ * A
= 224.81 MPa × 2000 $mm^2$
= 449.62 kN
Euler critical load
[13] $\lambda_0$ = 0.2 * sqrt(pi^2 * E / $p_y$) [1](Annex C)
= 0.2 × sqrt(π^2 × 205 GPa / 275 MPa)
= 17.155
Limiting slenderness
[14] η = max(0.001 * $a_R$ * (λ - $\lambda_0$), 0) [1](Annex C)
= max(0.001 × 5.5 × (94.868 − 17.155), 0)
= 0.42742
Perry factor (initial imperfection)

[15] $\varphi = (p_y + (\eta + 1) * p_E) / 2$
$= (275\ \text{MPa} + (0.42742 + 1) \times 224.81\ \text{MPa}) / 2$
$= 297.95\ \text{MPa}$
Perry–Robertson φ

[16] $p_c = p_E * p_y / (\varphi + \text{sqrt}(\varphi^2 - p_E * p_y))$ [1](Annex C)
$= 224.81\ \text{MPa} \times 275\ \text{MPa} / (297.95\ \text{MPa} + \text{sqrt}((297.95\ \text{MPa})^2 - 224.81\ \text{MPa} \times 275\ \text{MPa}))$
$= 133.78\ \text{MPa}$
Compressive strength

[17] $P_c = p_c * A$
$= 133.78\ \text{MPa} \times 2000\ \text{mm}^2$
$= 267.56\ \text{kN}$
Compression resistance

[18] $u = P / P_c$
$= 150\ \text{kN} / 267.56\ \text{kN}$
$= 0.56062$
Utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| r | 31.6 mm | Radius of gyration |
| λ | 94.868 | Slenderness |
| $p_E$ | 224.8 MPa | Euler stress |
| $P_E$ | 449.62 kN | Euler load |
| $\lambda_0$ | 17.155 | Limiting slenderness |
| η | 0.42742 | Perry factor |
| $p_c$ | 133.8 MPa | Compressive strength |
| $P_c$ | 267.56 kN | Compression resistance |
| u | 0.56 | Utilisation (governing) |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Axial load against compression resistance | u = 0.56 | u ≤ 1 | 0.56 | [1](Annex C) |

## 7. Applicability

None.

## 8. Verification

Template struct.column-buckling@1.0.1 Content hash b92d254599e2302a Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. British Standards Institution. Structural Use of Steelwork in Building—Part 1: Code of Practice for Design—Rolled and Welded Sections. BS 5950-1:2000. BSI; 2000.